

Predict Credit Card Default Data using Logistic Regression

For the Bachelor of Science Honours Degree in Financial
Mathematics and Industrial Statistics

By



SC/2020/11797
S.S.Liyanage

Supervisor
Ms. S.D.M.Dilshani

Department of Mathematics
University of Ruhuna
Matara
2024

Acknowledgment

First and foremost, we would like to express our sincere gratitude to Ms. S.D.M. Dilshani, our supervisor, for her tremendous advice and assistance during the planning and development of this case study. Her continuous feedback and feedforwards made our journey a success.

We are deeply grateful for the time and effort Ms. S.D.M. Dilshani has invested in reviewing and providing constructive feedback on this research proposal.

We are privileged to have had the opportunity to work under the mentorship of Ms. S.D.M. Dilshani, and I am confident that the knowledge and skills gained during this process will have a lasting impact on our academic and professional journey.

Also, our sincere gratitude goes to lecturers of the Department of Mathematics, for their support and guidance in keeping us progressing in conducting this case study.

We would like to extend our thanks to our colleagues, who provided valuable feedback and engaged in helpful discussions, enhancing the case study process.

Our special thanks go to every individual who is involved in this case study for their contributions and support throughout the journey.

Contents

Acknowledgment	2
1) Introduction	5
1.1) Objective of the Study	6
1.2) Research Question	6
2) Literature Review	7
2.1) Credit Card Defaults	7
2.2) Logistic Regression.....	7
2.2.1) AIC and BIC Values.....	7
3) Methodology.....	8
3.1)Data	8
3.1.1)Data source	8
3.1.2) Data Description.....	8
3.1.3) Data Cleaning	9
3.2) Methods.....	9
3.2.1) Logistic Regression.....	9
3.2.2) Multinomial logistic regression.....	10
3.2.3) Binary Logistic Regression	11
4) Results and Findings.....	14
4.1) Exploratory Analysis.....	14
4.2) Quantitative Analysis	18
5) Discussion and Conclusion.....	22
5.1) Discussion	22
5.2) Conclusion	22
References.....	23

Table of Figures

Figure 1 – Bar plot of variable student.....	14
Figure 2 – Bar plot of variable default.....	14
Figure 3 – Histogram of variable balance.....	15
Figure 4 – Histogram of variable income.....	15
Figure 5 – Correlation plot between balance and income.....	16
Figure 6 – Default status by income and balance.....	17
Figure 7 – Distribution of default by balance and income.....	17
Figure 8 – R code for data splitting.....	18

1) Introduction

The stability of a country's financial system has a direct impact on the state of the economy. The banking system is one of the key players in the financial system. The banking industry contributes a larger share of strength to the growth of the economy. In the modern financial environment, banks are leveraging new technology to enhance the quality of services they offer to their clients. With the ongoing evolution of innovative technologies, the world is moving to a cashless society. In such circumstances, credit cards play a major role as the mode of payment during transactions. A credit card can be identified as a small rectangular plastic card offered by a bank or a financial institute, but a definition of a credit card can be given as a kind of financial instrument that offers "revolving credit," a particular kind of loan. There is a credit limit on the loan, which may be utilized for both large and day-to-day purchases. (Hoffman, 2023)

The credit card is recognized as a competitive banking instrument that improves the bank's financial transactions and the financial position of the bank. Over the last ten years, credit cards have become the primary consumer lending instrument in Sri Lanka's banking sector. Additionally, using a credit card is the quickest and most flexible way to access bank funds on a short-term basis. People use credit cards to conveniently pay utility bills, make purchases, set up installment plans, and transact internationally. However, if the client fails to make the credit card payments on a monthly basis, the credit card will become delinquent, and it will negatively affect the bank as a result.

Defaulting on a credit card payment happens when a credit cardholder doesn't complete their obligation requirement to pay off their outstanding debt on time. Defaulting on a credit card commonly involves missing one or more scheduled payments, resulting in a series of consequences such as higher interest rates, late fines, and damage to the cardholder's credit score. It also damages the overall credit score of the bank as well. Predicting credit card default is a crucial part of risk management in the banking sector. In such a scenario, the powerful statistical modeling method logistic regression becomes especially useful since it provides information on the likelihood that a consumer defaulting on their credit card payments.

logistic regression, a regression analysis technique that is commonly applied when the dependent variable is categorical. Regarding the case of credit card default prediction, the outcome is binary and may be expressed as either "Yes" or "No" depending on whether the consumer defaults or not. In contrast to the continuous outcomes predicted by linear regression, logistic regression models the log odds of the probability that a given occurrence falls into a certain category. In logistic regression, the coefficients show how the dependent variable's log odds change when the corresponding independent variable changes by one unit. Evaluation measures that are used to evaluate the effectiveness of the logistic regression model in this study include AIC and BIC values. (Kleinbaum, 2010)

1.1) Objective of the Study

To develop a predictive model for credit card default using logistic regression with the aim of enhancing credit risk modeling techniques in the financial sector.

1.2) Research Question

“To study logistic regression technique to predict which customer will default on their credit card debt”

2) Literature Review

2.1) Credit Card Defaults

Failure to make at least the minimum payment on the outstanding balance on the credit card over an extended period of time is known as credit card default. A credit card default can be used to determine a customer's creditworthiness. (Wang, Wen, Zhou, & Zhang, 2023) For financial institutions throughout the globe, predicting credit defaults is an essential task. Banks and credit card firms must have precise models to predict credit failures due to the increase in the availability of financial goods and services and the rising number of credit users. (Lin, Nguyen, Lai, & Huang, 2023) According to Investopedia, revolving credit, or credit cards, allow borrowers to access money as required. Borrowers with revolving credit can access a predetermined amount of money up to a credit limit. However, you do not get that amount in full. Rather, you can spend the money in any way you see fit.

2.2) Logistic Regression

Logistic regression is a widely used statistical modeling technique in which the probability of a categorical outcome is related to a set of potential independent variables. (D.R. Cox, 1989) Logistic regression is a mathematical modeling technique that explains the relationship between one or more independent variables and one categorical dependent variable With the use of a linear function whose output shows the likelihood that an instance falls into one of the dependent variable's classes. (Agbemava et al., 2016) Lin et al (Lin, Nguyen, Lai, & Huang, 2023) Maximum likelihood estimate is used in logistic regression to find the ideal coefficients that characterize the connection between the independent variables and the outcome probability. These coefficients are commonly understood as the log odds or odds ratios, which show the effect of each independent variable on the likelihood that the result will occur.

2.2.1) AIC and BIC Values

Selecting the best model that balances model fit and complexity is a problem we frequently encounter when selecting models for regression analysis. The two widely used metrics for model selection in logistic regression are the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). Both AIC and BIC offer a quantifiable metric for assessing and contrasting various models, allowing data scientists to make well-informed decisions. AIC considers the probability, number of parameters, and sample size when weighing the trade-off between model fit and model complexity. However, BIC prioritizes simpler models by introducing a greater penalty for model complexity. (Kumar, vitalflux.com, 2023) According to (Chakrabarti & Ghosh, 2011) and (Kuha, 2004) the selection of statistical models is predicated on nested models' statistical significance testing. However, models may be rejected for having a statistically significant lack of fit because of unwanted features like sensitivity to small deviations from the null hypothesis. Due to similar problems with the significance tests have interested in alternate criteria. Two tools that are frequently used as model selection criteria are the Bayesian information criterion (BIC) and Akaike's information criterion (AIC).

3) Methodology

3.1) Data

3.1.1) Data source

The “Default” dataset is a generated dataset generally used in scientific and educational contexts to examine credit card default behavior. This dataset is a useful resource for researching the causes of credit card defaults and creating prediction models to support risk analysis and financial decision-making. This section provides a brief explanation of the given Default dataset, including the variables and data sources (Lin, Nguyen, Lai, & Huang, 2023).

The dataset is organized by the R program and also it is secondary data, which is typically sourced from the "ISLR" package (Introduction to Statistical Learning with Applications in R). This package issues datasets and functions that take place in the textbook "An Introduction to Statistical Learning" by James, Witten, Hastie, and Tibshirani.

According to the simulated nature of the dataset, researchers can examine the link between predictor factors and default probability in a controlled environment. To avoid missing any patterns or connections that might have a bearing on how the financial sector addresses risk assessment, the Default dataset is used in this study to predict credit card default. Since the dataset includes both binary and numeric variables, applying logistic regression to build a model to make the prediction is suitable.

3.1.2) Data Description

No	Variable Name	Type	Label Description
1	Default	Binary	A component that has "Yes" and "No" levels to indicate whether the customer has defaulted or not
2	Student	Binary	A component that indicates whether the customer is a student or not in levels of “Yes” and “No”
3	Balance	Numerical	The average amount that remains on a customer's credit card following their monthly payment
4	Income	Numerical	Income of the customer

The Default dataset includes 10,000 observations on credit card default data. The key variable of interest is "Default," a binary answer variable that indicates if a consumer defaulted on their credit card payment. It has two possible values: "No" (no default) and "Yes" (default). The dataset comprises predictor variables that might impact the chance of default, in addition to the default variable itself. Variables include customer information like student status "Student", credit card amount "Balance", and annual income "Income".

3.1.3) Data Cleaning

Using the statistical software R Studio, we completed relevant data pretreatment procedures, such as looking for missing values and outliers and analyzing exploratory data to guarantee the validity of our findings. To determine how well our predictive models performed, we further split the dataset into testing and training sections.

3.2) Methods

3.2.1) Logistic Regression

Logistic regression a "Supervised machine learning" approach is used to estimate the likelihood of a particular event. When there is a binary or dichotomous outcome and the data can be separated linearly, this method is employed. The most common use of logistic regression is in binary classification issues. (Bonthu, 2021) When the dependent variable indicates two categorical outcomes, binary logistic regression is applied. According to our study, we used binary logistic regression to build the model. In situations using multinomial or ordinal logistic regression methods, the dependent variable must include more than two variate outcomes.

Assumptions:

The dependent variable in binary logistic regression must be binary, and in ordinal logistic regression, it must be ordinal. And the observations must be independent of one another. It also requires the independent variables to have less or no multicollinearity. This indicates that there shouldn't be a high correlation between the independent variables. Further, log odds and independent variable linearity are assumed in logistic regression.

Pros and Cons:

Logistic regression is a very efficient training method and is simpler to use and analyze. It offers a direction of connection (positive or negative) in addition to a measure of a predictor's appropriateness (coefficient size). Model coefficients may be interpreted as measures of the significance of a feature. Although it is less likely to overfit, logistic regression can nevertheless overfit large dimensional datasets. There are cons as well. The presumption of linearity between the dependent and independent variables is the main limitation of logistic regression. Logistic regression does not apply to non-linear problems as it has a linear decision surface. In real-world scenarios, linearly separable data is rarely found. Average or nonexistent multicollinearity between independent variables is necessary for logistic regression.

The Logistic Regression Model:

$$\ln \left[\frac{P(Y)}{1 - P(Y)} \right] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

$$\frac{P(Y)}{1 - P(Y)} = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k}$$

$$P(Y) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k}}$$

Where, $\ln \left[\frac{P(Y)}{1 - P(Y)} \right]$ is the log odds of the outcomes, Y is the dichotomous

3.2.2) Multinomial logistic regression

Multinomial Logistic Regression is a variant of the logistic regression model that may predict outcomes from more than two categories. It is appropriate for scenarios in which the dependent variable contains three or more unorganized categories. The multinomial logistic regression model calculates the probability of each category relative to a reference category.

The model makes several parameter estimations, one for each category, compared to a selected reference category. These parameters show the change in the log odds of being in a certain category relative to the reference category for a one-unit range in the predictor variable. Logistic regression is an alternative to least-squares regression that ensures the fitted probabilities will be between 0 and 1. Group K, the final group, is chosen at random to act as the baseline category. In the model of multinomial logistic regression,

$$\log \left(\frac{P(y = k|x)}{P(y = K|X)} \right) = \beta_0^{(k)} + \beta_1^{(k)} x_1 + \dots + \beta_p^{(k)} x_p$$

For $k=1, \dots, K-1$

Generalized multinomial LRM:

$$\left(\begin{matrix} p(Y_{ik}) = p \\ (Y_i) = k \end{matrix} \middle| x_i; \beta_1, \beta_2, \dots, \beta_m \right) = \frac{e^{(\beta_{0k} + x_i \beta_k)}}{\sum_{j=1}^m e^{(\beta_{0j} + x_j \beta_j)}}$$

With $k=1, 2, \dots, m$ and β_m = row vector of regression coefficients of x for the kth category of y.

3.2.3) Binary Logistic Regression

Binary logistic regression is a strong statistical method for simulating the relationship between a binary dependent variable and one or more independent variables. Predicting the likelihood that an event will occur based on a collection of predictors is a widely used approach in many different sectors, including business, medical, and social sciences.

Assumptions:

The linear relationship between the independent factors and the dependent variable's logit transformation is one of the basic assumptions of binary logistic regression. This is based on the idea that the predictors and the log odds of the event happening have a linear relationship. A further assumption is that the independent variables are not multicollinear, which implies that their correlations should not be very strong. Multicollinearity can lead to unstable coefficient estimates and make it challenging to understand the individual impacts of the predictors.

Pros and Cons:

With its ability to include both categorical and continuous predictors in the analysis, binary logistic regression is a popular option for modeling binary outcomes. It is also a flexible tool for analyzing data with a variety of variable types because it provides estimates of odds ratios, which can be used to interpret the effects of the independent variables on the probability that the event will occur. The magnitude and direction of the relationships between the predictors and the outcome can also be understood with the help of these odds ratios. There are a few limitations with binary logistic regression that should be considered. one of its limitations is It assumes that there is a constant relationship between the predictors and the outcome at every level of the dependent variable

Binary logistic regression makes use of maximum likelihood estimation to determine the optimal coefficients that define the relationship between the independent variables and the probability of the outcome. These coefficients are typically interpreted as the log odds or odds ratios, which indicate the impact of each independent variable on the likelihood of the outcome occurring.

Theoretical framework:

In binary logistic regression, the dependent variable is modeled as a binary variable, typically denoted as "y," where $y = 1$ represents the occurrence of the event of interest (e.g., success) and $y = 0$ represents the absence of the event. The logistic function is defined as follows:

$$p = \frac{1}{1 + e^{-z}}$$

Where; p is the probability of the event of interest ($y = 1$); e = the base of the natural logarithm (approximately 2.71828), and z is the linear function of the predictors.

The linear function of the predictors is calculated as:

$$z = \log \frac{p}{1-p} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

Where; $\beta_0, \beta_1, \beta_2, \dots, \beta_p$ are the coefficients (also known as weights) associated with each independent variable; $x_1, x_2, \dots, x_p$ are the values of the independent variables; and p represents the number of independent variables (predictors) in the model.

The coefficients ($\beta_0, \beta_1, \beta_2, \dots, \beta_p$) in binary logistic regression are estimated using maximum likelihood estimation. The estimation process aims to find the values of the coefficients that maximize the likelihood of observing the given set of outcomes. Once the coefficients are estimated, they can be used to predict the probability of the event of interest for new observations by plugging in the corresponding values of the independent variables into the binary logistic function.

Based on the framework, we apply the model into our study and the model can be obtained as below,

$$\log \frac{Default}{1 - Default} = \beta_0 + \beta_1 * student + \beta_2 * balance + \beta_3 * income$$

Categorical variables could be somewhat more difficult to understand than continuous predictor variables, which add one independent variable to the equation. For instance, if a variable has four categories, then one of the categories becomes the base while the other three contribute three binary, independent variables that are mutually incompatible. As a result, we would analyze the variations in log odds that followed concerning the base category.

We will examine AUC, AIC, predicted accuracy, and weighted accuracy as performance indicators to assess the accuracy of these logistic regression models. The area under the ROC Curve (AUC) is a graph where the y- and x-axes represent the True Positive Rate (TPR) and False Positive Rate (FPR), respectively. The following is the formula for TPR and FPR, where FN and TN stand for false negatives and true negatives, respectively:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

Predicting true positives more accurately in the model will thus increase the area under the ROC and maximize the performance measure.

The Akaike Information Criterion (AIC) approximates the difference between the predicted model and a true model, so a lower AIC suggests better accuracy. We use this measure to compare the quality of each model against each other. The basic formula for AIC is

$$AIC = -2(\log(\text{likelihood of the fitted model}) + 2kk)$$

where log-likelihood represents the fit of the model and k represents the number of parameters in the model.

We will also compare predicted accuracy by calculating the proportion of credit cards that were accurately predicted to have been defaulted/not defaulted. However, the data set did not accurately reflect the actual distribution of defaulted credit cards. Weighted accuracy accommodates for this imbalance by putting more value in defaulted loans that are predicted accurately. The formula for calculating weighted accuracy is as follows:

$$\text{weighted accuracy} = \sum_{k=1}^{|G|} W_i \sum_{x:g(x)=k} I(g(x) = \widehat{g(x)})$$

We will also cross-validate our models to ensure that the model can adapt to different default credit data sets. Using a train-test split at 80:20 ratio will give the model enough data to train with while still leaving some for it to test with. We will also compare the models built with a null, or “coin toss,” model. This model randomly predicts defaults for credit cards based on the proportion of defaulted credit cards in the data set. Comparing the null model with other models will help us measure the impact of predictor variables.

4) Results and Findings

4.1) Exploratory Analysis

The default data set which has 10000 observations consists of two numeric variables, balance, and income, and two categorical variables default and student. A summary of the Default data set is given below.

default	student	balance	income
No :9667	No :7056	Min. : 0.0	Min. : 772
Yes: 333	Yes:2944	1st Qu.: 481.7	1st Qu.:21340
		Median : 823.6	Median :34553
		Mean : 835.4	Mean :33517
		3rd Qu.:1166.3	3rd Qu.:43808
		Max. :2654.3	Max. :73554

According to default status, there are 9667 observations where the customer did not default on the credit card and 333 observations where the customer defaulted on the credit card. Among the customers, there are 7056 observations where the customer is not a student and 2944 observations where the customer is a student. It is evident that among the customers most customers did not default on credit cards. In the case of balance, the minimum balance is zero and the maximum balance is 2654.3. The range of balance varies from 0.0 to 2654.3. In addition, 25% of the balances are below 481.7, 50% of the balances are below 823.6 and 75% of the balances are below 1166.3. Similarly, in the case of income, the range of income varies from 772 to 73554. 25% of the incomes are below 21340, 50% of the incomes are below 34553 and 75% of the incomes are below 43808.

The following bar plots represent the variables “default” and “student”.

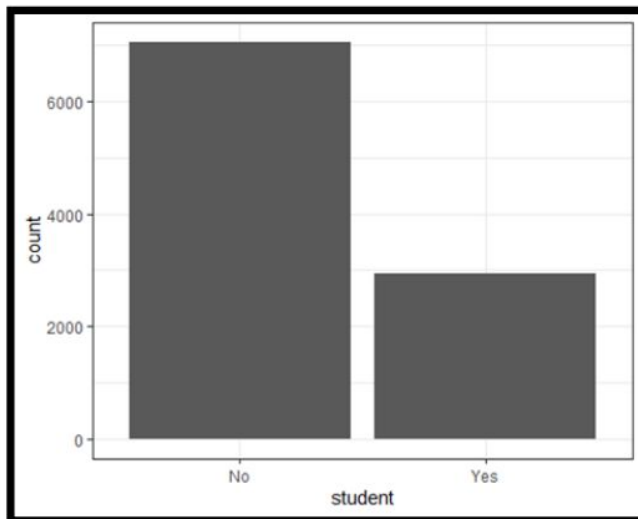


Figure 1 – Bar plot of variable student

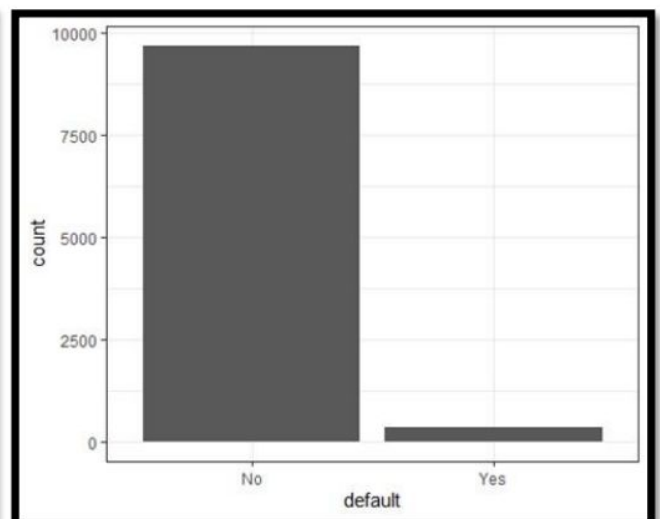


Figure 2 – Bar plot of variable default

Given below are the histograms of variable “balance” and “income”. A histogram is a powerful and versatile tool for analyzing, condensing, and visualizing the distribution of a dataset.

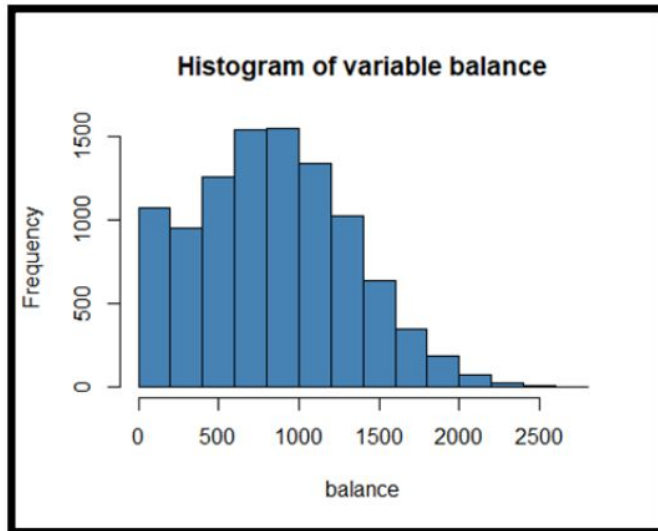


Figure 3 - Histogram of variable balance

Examining the above graph, it shows a right-skewed (or positive skewed) distribution. A common approach to describe a right-skewed graph is starting with symmetric and pulling its right tail. The majority of the observations are close to the lower end of the range and the higher observations are infrequent. According to the descriptive statistic of variable “balance,” the median is lower than the mean, which proves the right-skewed distribution. Since it shows skewness, we can conclude that variable balance is not normally distributed.

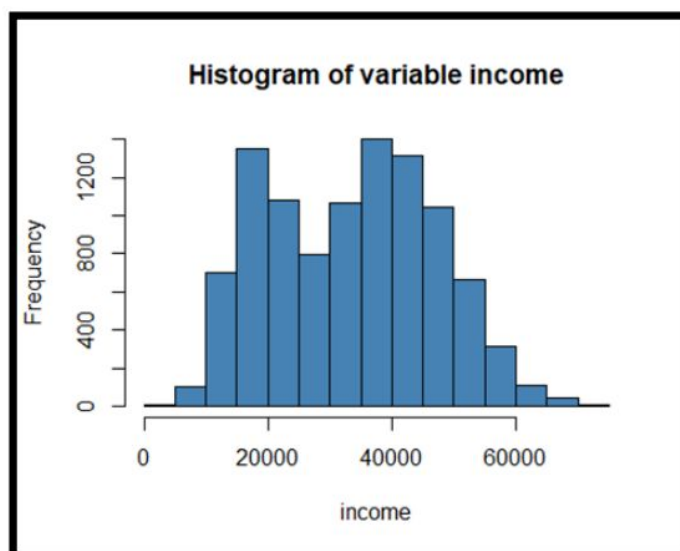


Figure 4 - Histogram of variable income

Histogram of variable “income” shows approximately bell-shaped curve, which the dataset is symmetric (the histogram looks roughly the same on either side of its center). It claims that the observations for variable “income” are normally distributed.

Correlation is a statistical tool used to assess how closely two variables change. A positive correlation indicates how much those variables rise or fall together. The range in which one variable rises while the other falls is indicated by a negative correlation. The following graph represents the correlation plot between variables “balance” and “income”.

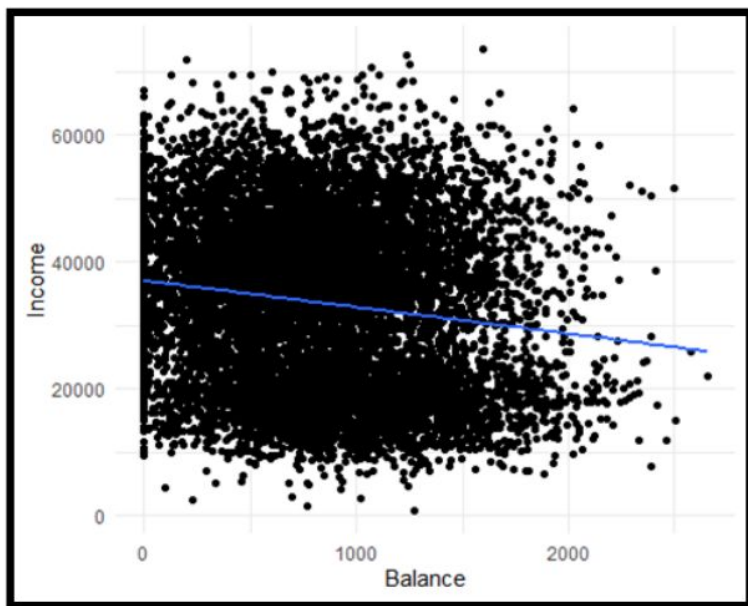


Figure 5 - Correlation plot between variables balance and income

The correlation coefficient obtained by R software is -0.1492335. It is evident that there is a negative correlation between variables “balance” and “income”.

Additionally, the following plot indicates the relationship between variables.

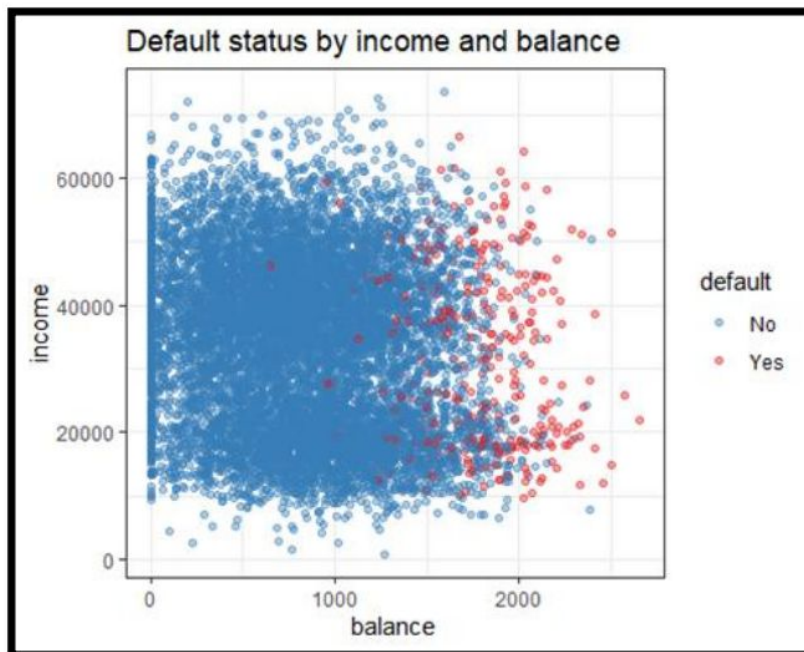


Figure 6 - Default status by income and balance

Through the graphical inspection, variable “balance” shows a finer predictor for variable “default” than the variable “income”.

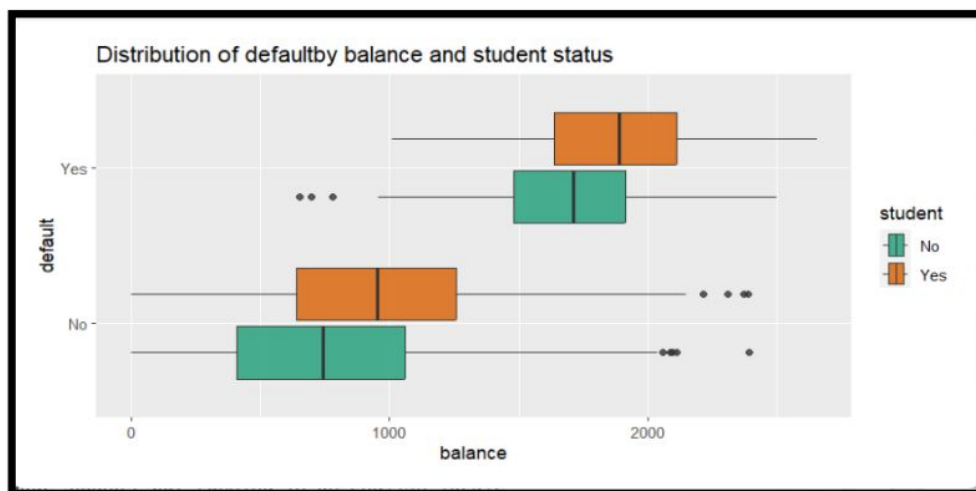


Figure 7 - Distribution of default by balance and income

Plot indicates that “balance” and “student” appear to be a reasonable indicator of “default”.

4.2) Quantitative Analysis

To generate the model, we split our data set into two parts according to proportion of 80/20, 80% of data as training data and 20% are testing data. In order to properly assess and train models, data dividing is a crucial first step in data analysis and machine learning . The procedure involves partitioning the given dataset into two or more subsets, a training set, and a testing set. The model is trained on the training set so that it can identify patterns and relationships in the data. On the other hand, the testing set is used to evaluate the model's effectiveness and its ability for generalization on untested data. Following is the code generated through R software to split the Default dataset.

```
##{r}
library(ISLR)
library(tidymodels)
set.seed(123)
# Split to Test and Train
Default_split <- initial_split(Default, prop = 0.8, strata = default)

Default_train <- training(Default_split) # 80% to training data
Default_test <- testing(Default_split) # 20% to test data

Default_split
```

Figure 8 - R code of data splitting

Step 1: To establish the beginning point for producing random integers, utilize the function "set.seed()." The sequence of randomly generated integers is guaranteed to stay the same each time the code is executed with the same seed value when a seed value is set using "set.seed()." Replicating results or producing reproducible analyses may benefit from this. Functions like sampling, permutation, and simulation that generate random numbers can have their randomness managed by setting a seed value. This enables the model to operate with unpredictable processes and produce consistent outcomes.

Step2: By default, the data is stratified sampling and divided into a 80/20 ratio. A technique called stratified sampling makes sure that the proportionate representation of various classes or categories in the data is preserved for both subsets. Using stratified sampling improves the generalizability and quality of the ensuing analysis or modeling procedures by producing training and testing sets that accurately reflect the original data's overall distribution and features.

Binary Logistic Regression Model:

Model is generated through 'glm' function in R software. Given below is the generated full model.

```
Call:
glm(formula = default ~ ., family = binomial, data = Default_train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.4850  -0.1383  -0.0538  -0.0195   3.7573

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.097e+01  5.583e-01 -19.642  <2e-16 ***
studentYes   -6.611e-01  2.655e-01  -2.490   0.0128 *
balance       5.802e-03  2.623e-04  22.121  <2e-16 ***
income       2.645e-06  9.262e-06    0.286   0.7752
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2320.3  on 7999  degrees of freedom
Residual deviance: 1229.4  on 7996  degrees of freedom
AIC: 1237.4

Number of Fisher Scoring iterations: 8
```

According to the above output our model is,

$$\log\left(\frac{Default}{1 - Default}\right) = \beta_0 + \beta_1 * student + \beta_2 * balance + \beta_3 * income$$
$$= -10.97 + 0.000000026 * income + 0.0058 * Balance - 0.0661 * student$$

Within this context, the variable "Student" is a categorical variable represented by a dummy variable, where "Yes" is coded as 1 and "No" is coded as 0.

According to that output, considering "Income" variable, the coefficient is determined to be 0.000000026, indicating a positive relationship between Income and log odds of Default's probability. It implies that for every 1 unit increase in Income, the log odds for the probability of default also increases by approximately 0.000000026 units. Therefore, this is providing that higher income levels are associated with a slightly higher likelihood of default.

According to "balance" variable, the coefficient is 0.0058. This interpret that for every 1 unit increase in the customers' balance, the log odds for likelihood of default also increase by 0.0058 units. It indicates a stronger impact of balance on the probability of default compared to the Income variable.

With respect to the “Student” variable, the coefficient is -0.0661. This negative coefficient suggests that customers who are students have a log odd for probability of default that is approximately 0.0661 units lower than customers who are not students. In other words, being a student is associated with a reduced likelihood of default, highlighting the potential influence of student status as a protective factor against default.

When examining the p-values, p-value is used in determining the significance of a variable in a model. The p-value of “income” variable is 0.7752 which is greater than 0.05. It indicates that variable “income” is not significant in explaining the variation in variable “default”. We can regenerate the model by removing “income” variable. The reduced model is given below.

```
Call:
glm(formula = default ~ student + balance, family = binomial,
    data = Default)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.4578  -0.1422  -0.0559  -0.0203   3.7435

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.075e+01  3.692e-01 -29.116 < 2e-16 ***
studentYes   -7.149e-01  1.475e-01  -4.846 1.26e-06 ***
balance       5.738e-03  2.318e-04   24.750 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2920.6  on 9999  degrees of freedom
Residual deviance: 1571.7  on 9997  degrees of freedom
AIC: 1577.7

Number of Fisher Scoring iterations: 8
```

When considering the p-values of the variables in the reduced model, both p-values of variables “student” and “balance” are less than 0.05 which indicates both the variables are significant in explaining the variation in variable “default”.

Model Selection:

The Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) are crucial metrics in model selection in regression analysis. Researchers can select the best model for their data with the use of quantitative measurements like AIC and BIC, which consider both model fit and complexity. AIC provides a balance between model complexity and goodness of fit, and BIC imposes a more severe penalty on it. Lower AIC and BIC values indicates the best fit model.

According to the credit card default data, a summary of AIC and BIC values of full model and the reduced model were generated through R. The following table indicates the summary of AIC and BIC values of reduced and full model.

A tibble: 2 × 3

Model <chr>	AIC <dbl>	BIC <dbl>
All Predictors	1237.431	1265.380
student + balance	1577.682	1599.313

2 rows

All predictors indicates the full model and student + balance indicates the reduced model.

Comparing AIC values:

Full model (All predictors) has the lowest AIC value of 1237.431 indicating a better fit compared to reduced model (student + balance) which has a AIC value of 1577.682.

Comparing BIC values:

Full model (All predictors) has the lowest BIC value of 1265.380 indicating a better fit compared to reduced model (student + balance) which has a BIC value of 1599.313.

Since full model (All predictors) has the lowest AIC and BIC values we can conclude that the full model is the best fit according to credit card default data.

5) Discussion and Conclusion

5.1) Discussion

This study discusses the components in logistic regression model, a kind of multivariable approach that is widely used to describe the connection between several independent factors and a categorical dependent variable. One of the most important parts of modeling in logistic regression involves selecting the variables. According to the credit card default data, when analyzing the logistic regression model for the full model, p-value of variable income was 0.7752 which was greater than 0.05. Following standard procedures to achieve statistical significance, the "income" variable was eliminated from the original model based on its non-significant p-value. After eliminating the variable "income", reduced model, which included the variables "student" and "balance," subsequently produced statistically significant relationships that highlighted the significance of both student status and credit card balance in predicting defaults.

However, when comparing AIC and BIC values to obtain the best fitted model between the full and reduced model, AIC and BIC values of the full model was recorded as 1237.431 and 1265.380 respectively. The less AIC and BIC values suggest that although variable "income" may not individually contribute to the statistical significance, its existence enhanced the model's overall explanatory power. Apparently, the full model which consist of variables "student", "balance" and "income" provides a solid basis for predicting credit card defaults. These findings contributes to a wide discussion on variable selection in logistic regression. Beyond the specific context of credit card default prediction, the study's findings emphasize the significance of a detailed approach to variable selection in logistic regression.

5.2) Conclusion

In conclusion, our case study about predicting credit card default data using logistic regression has provided insight into the relationship between model complexity and statistical significance. A customer's default status relies on three variables, income, balance, and student status indicating whether the customer is a student or not. The model indicates that all three variables have a significant influence on default likelihood. examining the relationship between these variables, according to Figure 1.6, It becomes clear that higher balances in accounts correspond to an increased likelihood of default status. Surprisingly, consumers with high incomes do not have a larger likelihood of default. additionally, Figure 1.7 indicates that customers who have student status appear to have a lower likelihood of default compared to non-student customers.

These results demonstrate the relationships that exist between income, balance, student status, and default likelihood, indicating that understanding and predicting default behaviors in credit cards depends on these factors. Further research is necessary to understand the underlying mechanisms driving these connections and improve the predictive models for more precise default predictions.

References

1. Lin, S.-H. *et al.* (2023) *Credit card default prediction: A comparative study of machine ...*, *Journal of Namibian Studies*. Available at: <https://namibian-studies.com/index.php/JNS/article/view/4580/3181> (Accessed: 26 January 2024).
2. Wang, Z. *et al.* (2023) *Credit card default prediction with Data Modeling*, *Atlantis Press*. Available at: <https://www.atlantis-press.com/proceedings/icbis-23/125989797> (Accessed: 26 January 2024).
3. Sirivongrungson, K. (2021) *Sign in*, *RPubs*. Available at: <https://rpubs.com/Lightbridge/809119> (Accessed: 26 January 2024).
4. Boateng, E.Y. and Abaye, D.A. (2019) *A review of the logistic regression model with emphasis on medical research*, *SCIRP*. Available at: <https://www.scirp.org/journal/paperinformation?paperid=95655> (Accessed: 26 January 2024).
5. Brownlee, J. (2020) *Probabilistic model selection with AIC, BIC, and MDL*, *MachineLearningMastery.com*. Available at: <https://machinelearningmastery.com/probabilistic-model-selection-measures/> (Accessed: 26 January 2024).
6. Rout, A.R. (2023) *Advantages and disadvantages of logistic regression*, *GeeksforGeeks*. Available at: <https://www.geeksforgeeks.org/advantages-and-disadvantages-of-logistic-regression/> (Accessed: 26 January 2024).
7. Bonthu, H. (2021) *An introduction to logistic regression*, *Analytics Vidhya*. Available at: <https://www.analyticsvidhya.com/blog/2021/07/an-introduction-to-logistic-regression/> (Accessed: 26 January 2024).

8. Peng, C., Lee, K.L. and Ingersoll, G. (2010) *An introduction to logistic regression analysis and reporting: ...* Available at: <https://www.tandfonline.com/doi/abs/10.1080/00220670209598786> (Accessed: 26 January 2024).
9. Arora, S. *et al.* (2021) *Prediction of credit card defaults through data analysis and ...* Available at: <https://sci-hub.se/downloads/2021-07-01/4e/arora2021.pdf> (Accessed: 26 January 2024).